
Software Requirements Specification

for

KKK Online System

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version
Chua Jia Lin Goe Jie Ying Lam Yoke Yu Tam Yi Ya Teh Ru Qian	3/11/2024	First Draft SRS	1.0
Chua Jia Lin Goe Jie Ying Lam Yoke Yu Tam Yi Ya Teh Ru Qian	15/11/2024	Scope changes and formatting	1.1
Chua Jia Lin Goe Jie Ying Lam Yoke Yu Tam Yi Ya Teh Ru Qian	29/11/2024	User password length changes	1.2

1. Introduction

Product Name: KKK Online System
Release Number: Version 1.0

This document specifies the software requirements for version 1.0 of the KKK Online System and covers all components of the system, providing requirements for the entire project scope.

1.1 Purpose

The purpose of this document is to guide the development of an online system designed to streamline the cooperative's member registrations, loan application processes and financial reporting. By digitising previously manual processes, the system aims to enhance data security, improve efficiency and accessibility for all stakeholders, including new applicants, members, administrators, and board members.

1.2 Document Conventions

This document uses the following conventions:

Fonts and Formatting	Chapter Names: 16-point Times New Roman in Bold Headings: 14-point Times New Roman in Bold Body Text: 12-point Times New Roman, justified
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1.3 Intended Audience and Reading Suggestions

This project serves as a prototype for the KKK Online System, developed for cooperative management within the organisation's framework. This project will benefit both the cooperative's management team and its members by enhancing access to services like member registration and loan applications.

1.4 Product Scope

This Software Requirements Specification (SRS) covers the entire KKK Online System, describing its features and functionalities for various stakeholders within the cooperative including new applicants, cooperative members, administrators, and board directors. It outlines:

1. New Member Registration

The new member can apply for cooperative membership using online application form, which reduces paperwork and enables remote access.

2. Loan Application Management

The loan application can be submitted through the online system, allowing applicants to track their application status online.

3. Administrator Controls

Administrators manage member data, prepare applications for board review, and update system policy.

4. Approval Process

The system allows applications to be printed for board meetings, where directors review and make approval decisions that administrators then enter into the system.

5. Financial Reporting

The system generates monthly and annual reports from stored data, aiding the cooperative in evaluating its financial and operational performance from time to time.

1.5 References

[1] Sommerville, I. (2016). "Software Engineering", 10th Edition, US: Pearson.

[2] "Portal Lembaga Kemajuan Pertanian Kemubu | KADA KEMUBU." <http://www.kada.gov.my/>

[3] R. Wherrity, "Waterfall Project Management: Comprehensive Guide For Success," Brand, <https://www.clearpointstrategy.com/blog/waterfall-project-management> (accessed Oct. 19, 2024)

2. Overall Description

This section contains description of the proposed system in various aspects including the product perspective, product functions, user classes and characteristics, operating environment, design and implementation constraints, user documentation, and assumptions and dependencies.

2.1 Product Perspective

This product is a standalone system developed to automate the manual processes of Koperasi Kakitangan Daerah Kelantan Berhad. Currently, the cooperative manages its operations including new member registration and loan application using manual methods such as paper forms and physical records. The proposed system aims to increase operational efficiency and reduce the cooperative workers burden as the cooperative processes can be done easily with the aid of the new system.

The system architecture can be explained using the Web Application Architecture diagram in Appendix B. The diagram shows users can reach the system using web browsers such as Microsoft Edge and Google Chrome, and the interfaces of the system are made using programming languages such as HTML and PHP. Besides, the system contains modules including membership application, loan application, and Generate Monthly and Annual Report. The database of the system is managed using MySQL database.

2.2 Product Functions

The major functions of the KKK Online System are as followed:

- **Log In:** Members can log in to the system using their username and password.
- **Apply Membership:** Non members need to apply for membership by filling the new member application form.
- **View Financial Status:** Members can view their financial status including the loan and share information.
- **Update User Profile:** Members can update their personal information and the family and inheritor information.
- **Apply Loan:** Members can apply for a loan by filling the loan application form.
- **View Loan Application Status:** The loan applicants can view their application status (pending, approved, or rejected).
- **Approve membership:** Admins can approve member applications of new users.
- **Approve loan application:** Admins can approve loan applications of members.
- **View list:** Admins can view member and loan applicant lists.
- **Record transaction:** Admins can record the transaction records and update financial status such as the amount of share capital and loan.

- **Update policies:** Admins can update policies such as the minimum membership fees and the financing rates.
- **Update Banner:** Admins can update advertising banners.
- **Generate Monthly and Annual Report:** Admins can generate and print monthly and annual financial reports.

The role of different user groups can be explained using the Use Case diagram in Appendix B. From the diagram, new users can only apply for membership using the system. For members, they can login to the system, update profile, apply loan and view application status, and view financial status. For admins, they can login to the system, Update policies and banners, record transactions, view lists for membership applicants and loan applicants, and generate reports.

2.3 User Classes and Characteristics

The proposed KKK Online System is designed mainly to support three user groups which are the new users, members of the cooperative, and the admins of the cooperative.

The new users should be able to do the following functions:

- Make membership application

The members of the cooperative should be able to do the following functions:

- Login to system
- Update profile
- Apply loan
- View loan application status
- View financial status through the member interface

The admins of the cooperative should be able to do the following functions:

- Login to system
- Record Transaction
- Approve membership application
- Approve loan application
- View members and loan applicants list
- Update policies
- Update advertising banner
- Generate and print monthly and annual financial reports

2.4 Operating Environment

The proposed system will work in a web-based environment, and is accessible through browsers such as Google Chrome, Firefox, Safari, and Microsoft Edge across desktop and mobile devices.

Minimum hardware requirements for **desktop**:

- **Operating System:** Windows 10, macOS Mojave, or Linux Ubuntu 20.04
- **Browser Compatibility:** Google Chrome (v100), Mozilla Firefox (v95), Safari (v13), Microsoft Edge (v100)

Minimum hardware requirements for **mobile devices**:

- **Operating System:** iOS 14 and Android 9.0
- **Browser Compatibility:** Google Chrome Mobile, Safari, and Firefox Mobile

2.5 Design and Implementation Constraints

The KKK Online System include several design and implementation constraints and they are as followed:

- **Security**
User authentication mechanisms such as user ID and password are crucial to prevent unauthorised access to the personal and financial data.
- **Performance**
The system should be able to handle multiple users especially during peak times by preventing system downtime or obvious lagging that will affect the cooperative operations.
- **Usability Testing**
Conduct usability testing with all user groups to ensure that the system functionalities and interfaces are user friendly and able to fulfil the users needs.
- **Backup and Disaster Recovery**
Create copies of data in real-time across different cloud locations, save daily backups, and prepare a recovery plan to prevent data loss if system downtime or disaster occurs.

2.6 User Documentation

The proposed system will include several types of documentation to guide users on how to use the system. These documentation are accessible in the system's Help section.

- **User Manuals:** Contains step-by-step guides for new users, members, and admins on how to use the system

2.7 Assumptions and Dependencies

The KKK Online System includes several assumptions and dependencies that may affect the operation of the system. They are as followed:

Assumptions:

- **User Adoption**
It is assumed that all user groups including new users, cooperative members, and admins are willing to change their manual processes to a digital system. Training sessions and user manuals will help them to adopt these changes.
- **Stable Internet Access**
Assume that all users will have a strong and stable internet connection that is important for a web-based system as any interruption could affect user experience and system functionality and efficiency.
- **Data Integrity**
Assuming the data being migrated from existing manual records to the digital system is accurate and complete, because disparity of data may cause application errors and user frustration.

External Dependencies:

- **Technological Compatibility**

The system's functionality is dependent on compatibility with standard browsers and mobile operating systems. If any of these platforms undergo significant changes such as a new version launched, the system may require updates to maintain compatibility.

3. External Interface Requirements

Interfaces will be included in this system to enable user interaction with the website.

3.1 User Interfaces

A web-based application would be the suggested method and it needs to have a user interface that is simple for both new and existing members. The logical features of the interface are as follows:

3.1.1 Login/Registration Page

1. **New user:** A membership application form that asks for personal information, employment information and supporting documentation from new users.
2. **Existing member:** A login page that allows users to safely access the system.

3.1.2 Member Dashboard

1. **Share Information (Maklumat Saham Ahli):** A section displaying the member's current share holdings.
2. **Loan Information (Maklumat Pinjaman Ahli):** Detailed view of the member's loan applications and outstanding amounts.

3.1.3 Administrator Interface:

1. **Application Management:** Administrators will be able to view and manage membership and loan applications.
2. **Report Generation:** Administrator can generate monthly and annual reports with filtering options.

3.1.3 Standard GUI Element

1. **Help and Support:** Every page has a help section that may be accessed from the header or footer.
2. **Error Handling:** Clear error messages displayed next to fields in forms and a standard error page for server-related issues.

3.2 Hardware Interfaces

The following hardware interfaces would be used to access the system:

1. **Client Devices:** Desktops, laptops and tablets capable of running modern web browsers.
2. **Web Server:** To interact with the client and host the application.
3. **Database Server:** To store application data, including user profiles, loan applications and financial data.
4. **Printing Interface:** The system's user interface and print command functionality is integrated.

3.3 Software Interfaces

The following software components are communicated with the system:

3.3.1 Frontend Technologies

1. **HTML:** Outlines the format and content of the web pages that users can interact with allowing for data presentation, form input and user navigation.
2. **CSS:** Styles the HTML framework to create a layout that is user friendly, ensuring a consistent look and feel across devices.
3. **JavaScript:** Improves the user experience by adding interactive elements including dynamic updates without page reloads, client-side form validation and responsive screen size adjustments.

3.3.2 Backend Technologies

1. **PHP:** Controls backend functions, including interacting with the database, processing form submissions, managing user authentication and producing dynamic content. PHP ensures real-time data interaction between the database and frontend.
2. **MySQL Database:** A relational database is used to store and manage cooperative data such as member and loan applicant records. MySQL will enable secured access to private data and provide safe data storage.

3.3.3 Online Web Hosting

All users will have remote access and high availability due to the system's cloud platform hosting. In order to provide continuous operation, cloud hosting improves scalability and provides backup and recovery options.

3.3.4 Email Service Integration

An email API will enable automatic notifications to users regarding their application status, updates and approvals. This will keep members informed and reduce manual communication.

3.4 Communications Interfaces

The following communication features will be required by the system:

1. **HTTP/HTTPS:** To make sure data encryption and safe communication, all web interactions between client and server will take place via HTTPS.
2. **Email Notifications:** Users will receive email notifications when they submit applications and when those applications are accepted or denied. The software's email API will be used to automatically generate these emails.
3. **Encryption:** Sensitive data, such as user personal information and financial information, will be encrypted both during transmission and in the database. Standards for encryption like RSA or AES-256 would be used.
4. **Synchronisation Mechanisms:** The system will use periodic triggers to synchronise with the database for backup operations and real-time updates, ensuring that no data is lost in case of network failure.
5. **Security Standards:** Encryption for all sensitive data communications, such as member financial information. Multi-factor authentication (MFA) as an added layer of security for both user login and admin access.

4. System Features

This section shows the major services offered by KKK Online System. This includes membership application, loan application, applicants, members and borrower listing and updating, reporting, record transaction, update policies, and update banner.

4.1 Membership Application

The cooperative operates for the mutual benefits of the members.

4.1.1 Description and Priority

The Member Application feature allows users to apply for the membership of the cooperative. The user would have to fill out the online form and upload any required documents.

- **Priority:** High
- **Benefit:** 9 (High)
- **Penalty:** 8 (High)
- **Cost:** 5 (Medium)
- **Risk:** 3 (Low)

4.1.2 Stimulus/Response Sequences

1. **User Action:** User clicks “Permohonan Menjadi Anggota” in the log in page.

System Response: Display the registration form, section “Maklumat Pemohon”.

2. **User Action:** User fills in the form and clicks “Simpan”.

System Response: Validate input; if valid, create a member application profile and display a prompt indicating the information is saved. If invalid, prompt error messages.

3. **User Action:** User clicks other sections of the form, namely “Maklumat Keluarga / Pewaris”, “Yuran dan Sumbangan” and “Akuan Kebenaran” or “Seterusnya”.

System Response: Display the selected section of the registration form.

4. **User Action:** User fills in the form and upload relevant documents.

System Response: Validate input; if valid, update the application profile. If invalid, prompt error messages.

5. Repeat steps 3 and 4 until all sections of the form are completed and validated.

6. **User Action:** User clicks “Seterusnya” at the “Akuan Kebenaran” page.

System Response: System displays “Semakan Butiran”.

7. **User Action:** User checks the information and clicks “Hantar”.

System Response: Prompt status for user and send an email to the user and the administrator respectively.

4.1.3 Functional Requirements

REQ-1: The system shall validate the provided email address format, identity card number and phone number.

REQ-2: The system shall display error messages for invalid input, such as missing

required fields or incorrect input format

REQ-3: The system shall send an email to the user and the administrator upon the submission of the application.

REQ-4: The system shall ensure that all member data is stored securely and comply with relevant data protection regulations.

4.2 Loan Application

The cooperative offers financing services for their members.

4.2.1 Description and Priority

The Loan Application feature allows users to apply for the financing from the cooperative. The applicant must be already a member of the cooperative and fulfil the minimum share capital requirement of RM 300.00. The user would have to fill out the online form and upload any required documents.

- **Priority:** High
- **Benefit:** 9 (High)
- **Penalty:** 8 (High)
- **Cost:** 5 (Medium)
- **Risk:** 3 (Low)

4.2.2 Stimulus/Response Sequences

1. **User Action:** Use clicks “Pinjaman”

System Response: Display the registration form, section “Butir-Butir Pembiayaan”.

2. **User Action:** User fills in the form and clicks “Simpan”.

System Response: Validate input; if valid, create a loan application profile and display a prompt indicating the information is saved. If invalid, prompt error messages.

3. **User Action:** User clicks other sections of the form, namely “Butir-Butir Peribadi Pemohon”, “Pengakuan Pemohon”, “Butir-Butir Penjamin”, “Pengesahan Majikan” and “Akuan Kebenaran” or “Seterusnya”.

System Response: Display the selected section of the registration form.

4. **User Action:** User fills in the form and upload relevant documents.

System Response: Validate input; if valid, update the application profile. If invalid, prompt error messages.

5. Repeat steps 4 and 5 until all sections of the form are completed and validated.

6. **User Action:** User clicks “Seterusnya” at the “Akuan Kebenaran” page.

System Response: System displays “Semakan Butiran”.

7. **User Action:** User checks the information and clicks “Hantar”.

System Response: Prompt status for user and send an email to the user and the administrator respectively.

4.2.3 Functional Requirements

* REQ-2, REQ-3 and REQ-4 from the Membership Application are also applicable to

Loan Application.

REQ-5: The system shall retrieve and auto-fill relevant user information in the "Butir-Butir Peribadi Pemohon" section based on the user's profile, so that the user does not need to re-enter the same information.

4.3 Applicants, Members and Borrowers Listing and Updating

The administrator should be able to view the list of members who apply the membership and loan, and the list of existing members and members who are under loan.

4.3.1 Description and Priority

The Board of Directors of the cooperative will conduct monthly meetings to review the application to approve or reject the member and loan applications. After the meeting, the administrator will update the application status according to the result of the meeting.

- **Priority:** High
- **Benefit:** 8 (High)
- **Penalty:** 7 (High)
- **Cost:** 5 (Medium)
- **Risk:** 3 (Low)

4.3.2 Stimulus/Response Sequences

1. **User Action:** Admin clicks "Permohonan Anggota" or "Permohonan Pinjaman."

System Response: Display the list of membership applications or loan applications.

2. **User Action:** Admin clicks the status "Sedang Diproses".

System Response: System displays a dropdown list of "Diterima", "Ditolak" and "Sedang Diproses".

3. **User Action:** Admin clicks a choice from the dropdown list.

System Response: Update the application and related databases. An email is sent to the applicant to notify the applicant about the application status.

4. **User Action:** Admin clicks "Senarai Anggota" or "Senarai Pinjaman."

System Response: Display the list of members or borrowers.

5. **User Action:** Admin clicks a member from the list.

System Response: System displays the information of the member or the information of the financing.

4.3.3 Functional Requirements

REQ-6: The system shall display the list of members and their information.

REQ-7: The system shall update the application status in the database upon selection by the administrator.

REQ-8: The system shall send an email notification to the applicant regarding the updated application status after the administrator makes the selection.

REQ-9: The system shall maintain an audit log of status changes made by the administrator, including timestamps and admin ID.

REQ-10: The system shall display a confirmation message to the administrator after successfully updating the database.

4.4 Reporting

A report can be generated to view the members and financial summary.

4.4.1 Description and Priority

The administrator can obtain the monthly and annually account report of the cooperative.

- **Priority:** High
- **Benefit:** 9 (High)
- **Penalty:** 6 (Medium)
- **Cost:** 6 (Medium)
- **Risk:** 4 (Medium)

4.4.2 Stimulus/Response Sequences

1. **User Action:** Admin clicks “Laporan”.

System Response: Display two dropdowns namely “Bulanan” and “Tahunan”.

2. **User Action:** Admin selects and clicks the month or year.

System Response: System displays the report of the selected month or year.

4.4.3 Functional Requirements

* REQ-9 and REQ-10 from the *Applicants, Members and Borrowers Listing and Updating* are also applicable to Reporting.

REQ-11: The system shall generate and display the report for the selected month or year upon selection by the administrator.

REQ-12: The system shall allow the administrator to export the report in various formats (e.g., PDF, Excel).

4.5 Record Transaction

Transactions such as share purchasing, salary deduction and loan repayment are done manually. Thus, the administrator would need to update the transaction made.

4.5.1 Description and Priority

This feature allows the administrator to update transactions that may affect the user's financial status. This includes updating the amount of share capital, funds, contributions and loans.

- **Priority:** High
- **Benefit:** 9 (High)
- **Penalty:** 8 (High)
- **Cost:** 4 (Medium)
- **Risk:** 3 (Low)

4.5.2 Stimulus/Response Sequences

1. **User Action:** Admin clicks “Transaksi Baru”.

System Response: Display the list of members.

2. **User Action:** Admin searches the member name.

System Response: System the member.

3. **User Action:** Admin clicks on the member.

System Response: System displays the financial status of the member and fields for the admin to update the amount.

4. **User Action:** Admin enters the amount to be added or subtracted.

System Response: System prompts a confirmation message.

5. **User Action:** Admin confirms the action.

System Response: System updates the database.

4.5.3 Functional Requirements

* REQ-9 and REQ-10 from the *Applicants, Members and Borrowers Listing and Updating* are also applicable to *Reporting*.

REQ-13: The system shall display the current financial status of the member as the administrator selects the respective member.

REQ-14: The system shall validate the input fields for policies and ensure that the amount does not exceed the limit or form a negative record.

REQ-15: The system shall update the related databases with the new financial status upon confirmation of the administrator.

4.6 Update Policies

The economical situation would fluctuate and the interest rate would change. Besides, the requirements would change in the long run.

4.6.1 Description and Priority

This feature allows the administrator to update policies such as the minimum membership, minimum share capital requirement for loan application, loan interest rate, maximum loan repayment duration and maximum financing amount. This ensures the sustainability of the system.

- **Priority:** Medium
- **Benefit:** 8 (High)
- **Penalty:** 6 (Medium)
- **Cost:** 5 (Medium)
- **Risk:** 3 (Low)

4.6.2 Stimulus/Response Sequences

1. **User Action:** Admin clicks “Kemaskini Polisi” or “Kemaskini Iklan”.

System Response: Display the list of policies or the banner.

2. **User Action:** Admin clicks “Kemaskini”.

System Response: System displays a form.

3. **User Action:** Admin change the value of the form or update a new image for the banner.

System Response: Update the application and related databases.

4.6.3 Functional Requirements

* REQ-9 and REQ-10 from the *Applicants, Members and Borrowers Listing and Updating* are also applicable to *Update Policies*.

REQ-16: The system shall display the current list of policies when the administrator selects the respective option.

REQ-17: The system shall validate the input fields for policies.

REQ-18: The system shall update the application and related databases with the new policy upon submission by the administrator.

4.7 Update Banner

The banner in the member main page would display some notice or promotions. The banner would change from time to time and requires update.

4.7.1 Description and Priority

This feature allows the administrator to update the website banner. This ensures the sustainability of the system.

- **Priority:** Medium
- **Benefit:** 8 (High)
- **Penalty:** 6 (Medium)
- **Cost:** 5 (Medium)
- **Risk:** 3 (Low)

4.7.2 Stimulus/Response Sequences

4. **User Action:** Admin clicks “Kemaskini Polisi” or “Kemaskini Iklan”.

System Response: Display the list of policies or the banner.

1. **User Action:** Admin clicks “Kemaskini”.

System Response: System displays a form.

2. **User Action:** Admin change the value of the form or update a new image for the banner.

System Response: Update the application and related databases.

4.7.3 Functional Requirements

* REQ-9 and REQ-10 from the *Applicants, Members and Borrowers Listing and Updating* are also applicable to *Update Banner*.

REQ-19: The system shall display the existing banner when the administrator selects the respective option.

REQ-20: The system shall validate the input fields for policies and ensure that uploaded images meet specified size or format criteria.

REQ-21: The system shall update the application and related databases with the new banner information upon submission by the administrator.

REQ-22: The system shall display a confirmation message to the administrator after successfully updating the policies or banner.

5. Other Nonfunctional Requirements

This section defines the constraints on the services and functions offered by the KKK system. Nonfunctional Requirements includes Performance Requirements, Safety Requirements, Security Requirements, Software Quality Attributes and Business Rules.

5.1 Performance Requirements

This section defines the software system response time of user actions under a certain workload, throughput and resource usage.

- 5.1.1 System shall be able to maintain 99.9% uptime, making it always available for corporate users. Service interruption or temporal shut down such as maintenance or update should be minimised or communicated beforehand.
- 5.1.2 System response time should be at most 1 seconds under normal loads.
- 5.1.3 System should be able to support at least 100 active user sessions at the same time and multiple user sessions should not slow down response time.
- 5.1.4 The system must retrieve member data within 1 second of a search query under normal load with up to 50 concurrent queries.

5.2 Safety Requirements

This section specifies the requirements that are concerned with possible loss, damage or harm that could result from the usage of the system.

- 5.2.1 System should ensure all personal and financial data collected during membership registration or loan applications is protected against unauthorised access by the General Data Protection Regulation (GDPR) and ISO/IEC 27001 standards.
- 5.2.2 All activities should be logged and able to retrace back, to provision recovery from system failure All logs must be protected against unauthorised changes, and can only be accessed by authorised personnel for maintenance purposes.
- 5.2.3 The system must be capable of restoring itself to the nearest previous state if an error occurs, for example system crash or a connectivity loss.

5.3 Security Requirements

This section states the requirements regarding any security or privacy concerns surrounding use of the system and protection of the data used or collected. User identity authentication requirements are defined.

- 5.3.1 The system must have an automated data backup and recovery mechanism to prevent data lost. Regular backup should be scheduled and backup data are stored in a secure, offsite location.
- 5.3.2 Passwords of all users of the system must have a bit strength of at least 64 bit, that is at least 6 characters long and includes a mix of uppercase and lowercase letters, numbers, and special symbols.
- 5.3.3 Wireless communication throughout the system will be encrypted using TLS 1.3 at the application layer and WPA2-Enterprise at the data link layer.
- 5.3.4 The system shall provide two level of access:
 - An administrator level for unrestricted access to all system functionality.
 - A member layer for access to member functionality only.

5.4 Software Quality Attributes

Additional quality characteristics for the product that will be important to either the customers or the developers are defined in this section.

- 5.4.1 System must be compatible with different operating systems such as Windows, macOS, Linux, and mobile devices like iOS and Android. System must be adaptable with all device types, including workstations, laptops, tablets and smartphones.

- 5.4.2 The system must fully meet its requirements specifications, with any deviation from the requirements considered as incorrect.
- 5.4.3 Systems should be reliable, failures should be minimised and issues that could lead to software failure should be addressed.
- 5.4.4 System should be easy to use, be learnable to beginners, convenient, and effective in handling errors.

5.5 Business Rules

This section defines the policies and constraints the system must adhere to, and functions specific to roles that aligns with the organisational practices.

- 5.5.1 Only the admin can modify loan repayment terms and policies.
- 5.5.2 Accessing member personal information is limited to admin authority. System should restrict access to member data by non admin role, except every member is able to view and modify their own personal data.

6. Other Requirements

6.1 Localisation Requirements

The system is used exclusively for the KKK, and the cooperative is in Malaysia.

- Languages supported: Malay only.
- Date formats: DD/MM/YYYY
- Time zone adjustment: UTC+8
- Currency: Ringgit Malaysia (RM)

6.2 Data Migration Requirements

The KKK Online System is replacing the existing manual system. Thus, an interface is required for the admin to add existing data to the new system.

Appendix A: Glossary

Acronyms	Meaning
KKK	Koperasi Kakitangan KADA Kelantan Berhad
KADA	Lembaga Kemajuan Pertanian Kemubu
SRS	Software Requirements Specification
HTML	HyperText Markup Language
PHP	Hypertext Preprocessor
user ID	User identification
GUI	Graphical User Interface
CSS	Cascading Style Sheets
API	Application programming interface
HTTP	Hypertext Transfer Protocol
RSA	Rivest–Shamir–Adleman
AES	Advanced Encryption Standard
REQ	requirement
ISO/IEC	International Organization for Standardization/International Electrotechnical Commission
TLS	Transport Layer Security
WPA	Wi-Fi Protected Access
UTC	Coordinated Universal Time

Appendix B: Analysis Models

Web Application Architecture diagram for KKK Online System:

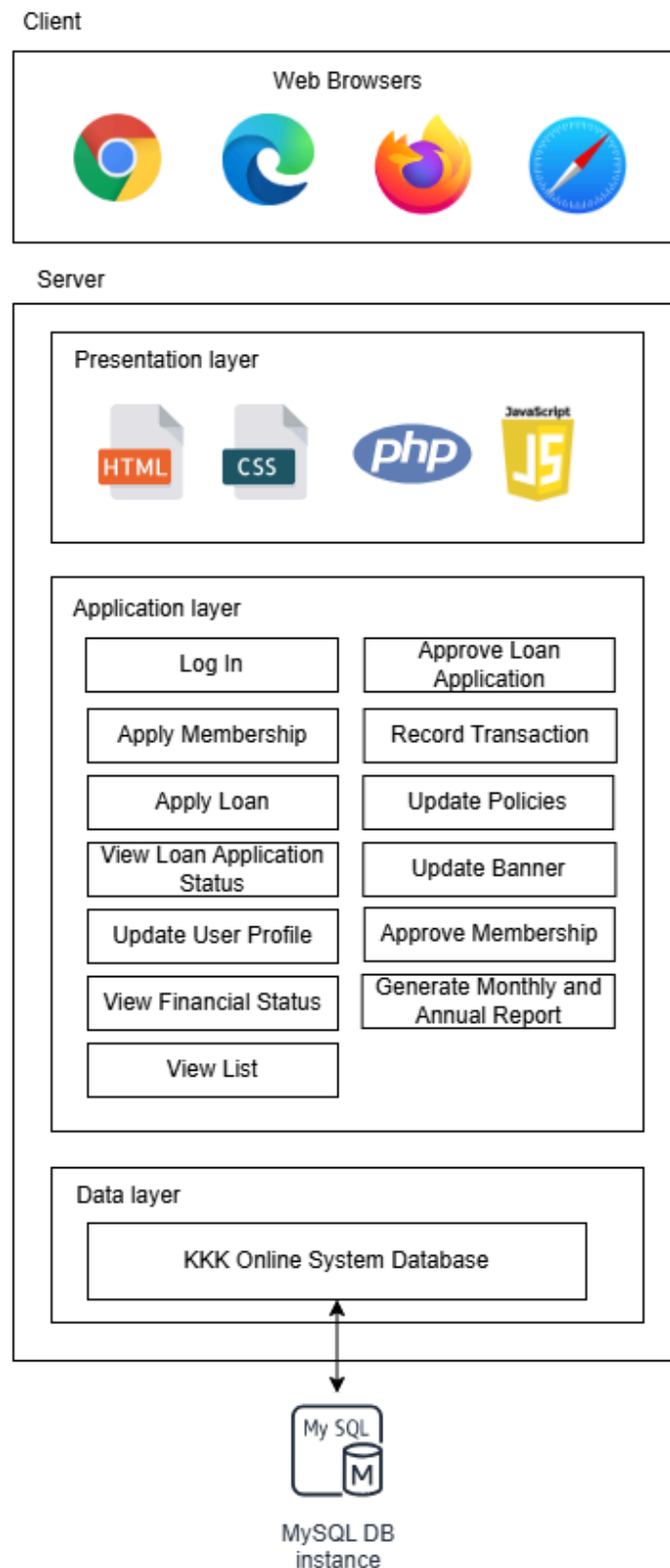


Figure 1 Web Application Architecture diagram

Use Case diagram of KKK Online System:

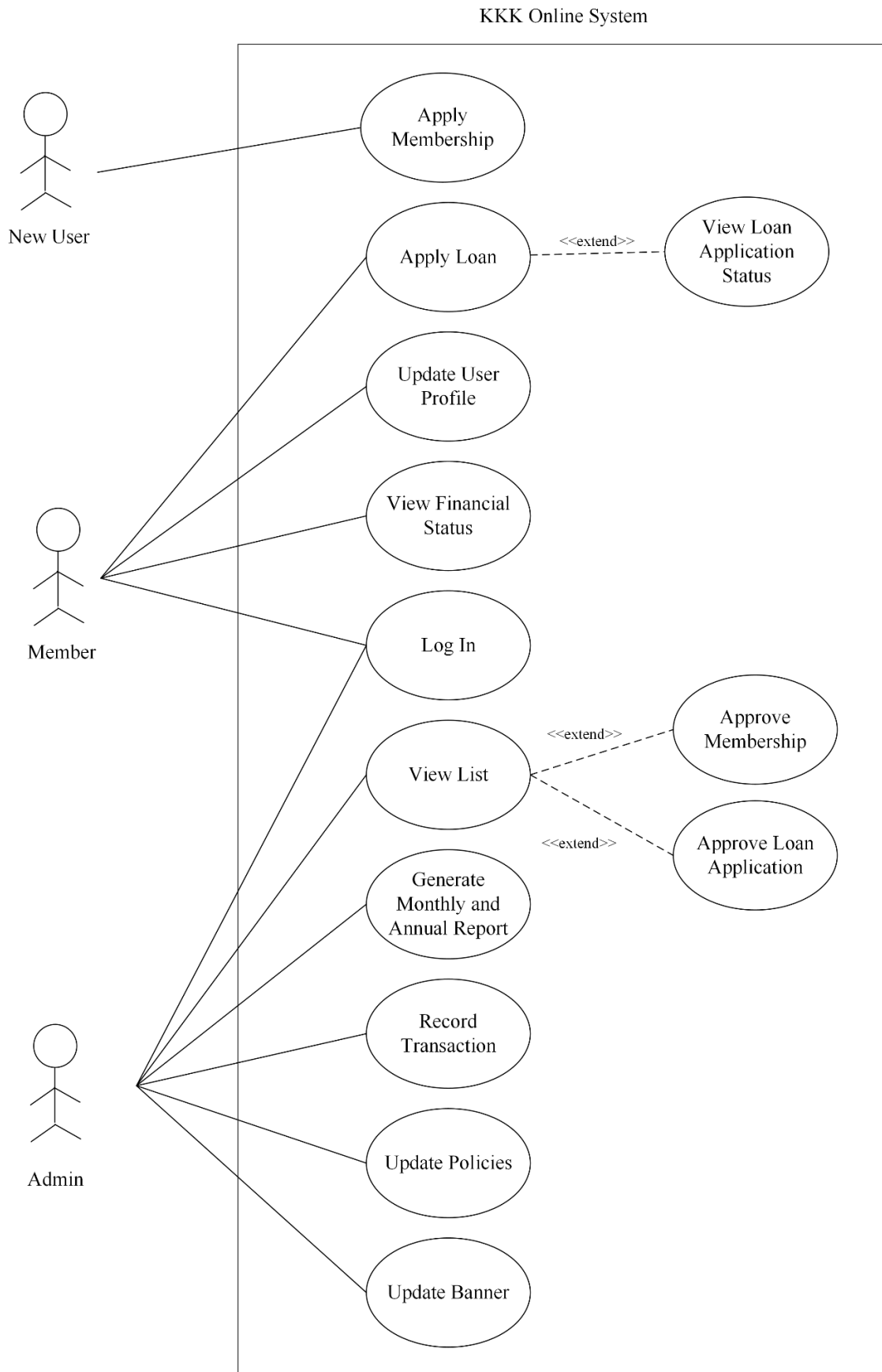


Figure 2 Use Case Diagram for KKK Online System